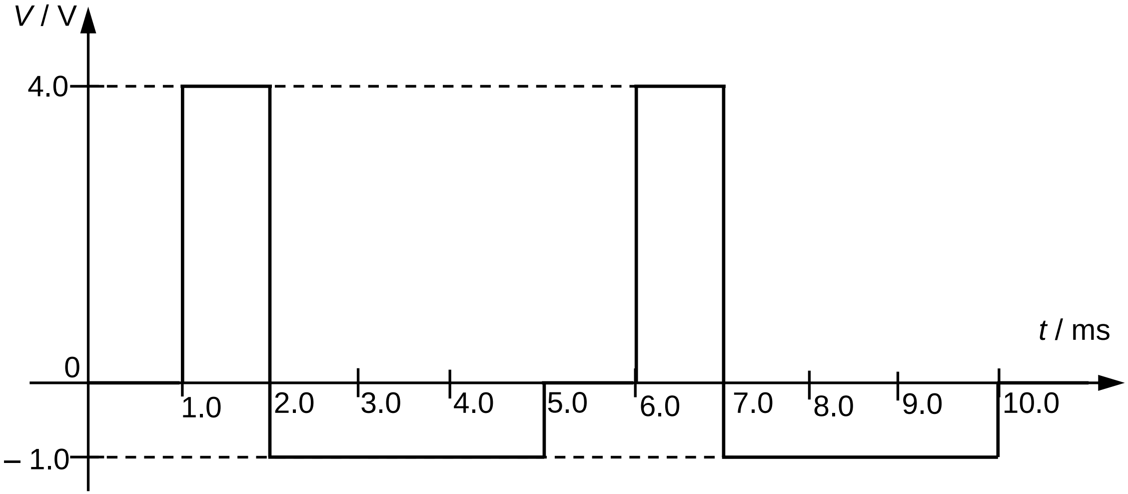
27 The variation with time t of the voltage V of an alternating source applied across a 2.5 Ω resistor is shown below.



What is the mean power dissipated in the resistor?

[Turn over

	A 0.98 W	**B** 1.5 W	**C** 1.9 W	**D** 2.4 W

Answer: B

To find r.m.s. value of V, we need to square the graph and find the average value of V^2 then find square root.

$$V_{r.m.s.} = \sqrt{\frac{(4^2 \times 1) + (1 \times 3)}{5}} = \sqrt{(3.8)}$$

Power dissipated = $V_{r.m.s.}^2$ / R = 3.8 / 2.5 = 1.5 W

28	The graph below shows the variation of X-ray intensity with wavelength emitted from an